

Week 5 Day 4

Comparison Report:

Sequential vs. Hierarchical vs. Single-Agent

Introduction

This report compares three approaches used to complete the laptop recommendation task: a Single-Agent LangGraph workflow, a Sequential CrewAI workflow, and a Hierarchical CrewAI workflow. The comparison focuses on output quality, execution cost, and overall effectiveness.

Single-Agent Workflow

The single-agent workflow completed the entire task independently. It achieved good quality while consuming the fewest tokens, making it the fastest and most cost-efficient approach for a relatively straightforward recommendation task.

Sequential Crew Workflow

The sequential workflow assigned research, analysis, and communication to separate specialist agents. This produced accurate, organized, and customer-friendly results with moderate token usage. The division of responsibilities improved the structure and reliability of the final recommendation.

Hierarchical Crew Workflow

The hierarchical workflow introduced a manager agent responsible for coordinating specialist agents. During testing, delegation issues prevented the workflow from executing correctly, resulting in hallucinated product information and significantly higher token consumption. We re-run, the model quota reached its limit. Although hierarchical coordination can be valuable for complex projects, it added unnecessary overhead for this task.

Cost and Quality Comparison

Approach	Quality	Cost	Speed	Notes
Single-Agent (LangGraph)	Good	Low	Fast	Handled the task well with the lowest token usage. Best option for straightforward product recommendations where only one reasoning step is needed.
Sequential Crew	Very Good	Moderate	Moderate	Each agent had a clear responsibility (research, analysis, communication), producing structured and accurate results. More expensive than a single

				agent, but outputs were easier to control and verify.
Hierarchical Crew	Mixed	High	Slow	Intended to let the manager coordinate the specialist agents, but delegation issues prevented it from working correctly. It consumed the most tokens because of the manager's planning, delegation attempts, and additional reasoning overhead.

Conclusion

Overall, the **Sequential Crew** provided the best balance between output quality and resource usage. The Single-Agent solution remained the most economical option, while the Hierarchical Crew demonstrated the highest operational cost without delivering additional value for a relatively simple recommendation task.